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Exploring Regional Profile of Drought History- a New Procedure to Characterize and Evaluate Multi-Scaler Drought Indices Under Spatial Poisson Log-Normal Model

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Abstract
Drought is recurrently occurring in many parts of the globe. In contrast to other natural hazards, drought has complex climatic characteristics. Several environmental factors are involved in the occurrence of drought hazards. However, the selection of an appropriate drought index may incorporate to make efficient drought mitigation policies. Moreover, the spatial distributions of extreme weather conditions (Dry/Wet) are necessary to avoid the consequences of future drought hazards. In this study, we aimed to explore and compare the regional profile of Dry/Wet episodes under various drought measures. Consequently, this research proposes a new spatial comparative procedure to assess and evaluate the spatial predictive distributions of Dry/Wet episodes under various drought measures. The study incorporates three Standardized Drought Indices (SDIs) and a spatial Poisson log-normal model to assess and evaluate the spatial predictive distributions of Dry/Wet episodes in Pakistan. Results of this study show that the segregated patterns of Dry/Wet counts are moderately consistent with the climatology of the region. However, the spatial patterns of Wet/Dry counts under various drought measures are significantly different under each index. Therefore, this research suggests the simultaneous use of multiple drought measures for accurate and precise drought monitoring at the regional level.

Keywords Drought · Standardized drought indices · Spatial Poisson log-normal model · Spatial predictive distribution

1 Introduction

Dynamic climatology and global warming have affected the spatiotemporal patterns, amount, and cyclic variation in rainfalls since the industrial revolution (Nam et al. 2016). On the other hand, the water demand has increased due to the expansion of the agriculture and industrial sectors. Consequently, human lives, agricultural activities, energy sectors, livestock, wildlife, and ecosystems have been adversely affected across the globe

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(Wilhite et al. 1993). Apart from other consequences, the primary feature of global warming is drought hazards. Drought is a most disastrous and complex natural hazard (Dracup et al. 1980). Unlike other natural hazards, it occurs in many parts of the globe. However, its characteristics like intensity, duration, frequency, and spatial extent vary from region to region (Zhou et al. 2020). Overall, climate change has increased the duration and the magnitude of drought in many regions around the globe (Tigkas et al. 2020; Chiang et al. 2021).

To avoid the adverse impacts of drought, accurate and reliable information regarding incoming extreme events (i.e., drought and flood) is inevitable. Since multiple climatic factors are responsible for creating drought phenomena, it is necessary to obtain accurate and reliable information on future droughts (Kiem et al. 2016).

Numerous researchers have proposed various drought monitoring tools for various regions. Among these tools, Standardized Drought Indices (SDIs) such as the Standardized Precipitation Index (SPI), Standardized Precipitation Evapotranspiration Index (SPEI), and Reconnaissance Drought Index (RDI) (Erhardt and Czado 2015; McKee et al. 1993; Vicente-Serrano et al. 2010; Tsakiris et al. 2007) are among the most widely used and internationally accepted methods. In several studies, SDIs type indicators have been used to forecast drought under various probabilistic and deterministic techniques (Dikshit et al. 2021). For example, Mishra and Desai (2005) utilized various stochastic models, including Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA), to forecast quantitative values of the Standardized Precipitation Index (SPI) drought index. Recently, Xu et al. (2020) forecasted the SPI drought index using the Hybrid Autoregressive Integrated Moving Average- Support Vector Regression (HARIMA–SVR) model. Zargar et al. (2011) have provided a review of the list of drought indices with their structures for drought monitoring.

However, the climatic variables and methodologies of each indicator in SDI modules are varied from region to region as they depend upon the availability of data and the nature of the environment. Further, the subjective choice of SDIs type indicators creates a chaotic situation for global climate policies. Therefore, evaluating drought indices at a regional level is essential for drought monitoring and early warning risk assessment.

Further, sufficient information about the prevalence of drought at the regional scale is required to establish effective drought mitigation policies. In recent research, several studies investigated drought spatial and spatiotemporal patterns using geostatistical methods (Wolski et al. 2021; Jiang et al. 2022; Ji et al. 2021). However, the difference between the frequencies of extreme climatic conditions, such as the dynamic structure of the rainfall pattern, is the most challenging issue around the world. This research aims to explore and compare the regional profile of Dry/Wet episodes under various drought measures. Consequently, this paper provides a new comparative assessment procedure for comparing multiscaler drought indices at the regional level.

The distribution of this paper is as follows: Sect. 2 summarizes the stepwise structure of the proposed approach for analyzing the regional profile of drought and spatial comparison of drought indices. Section 3 consists of materials and methods. Section 4 describes the exploratory spatial analysis. Results are presented in Sect. 5. Finally, Sects. 6 and 7 provide discussion and conclusion, respectively.

2 The Outlines of the Proposed Procedure

This section summarizes the proposal of the research for comparing various multi-scaler drought indices. In Fig. 1, flowchart of the proposed procedure is presented. The following lines provide the stepwise execution of the proposed method.

1. **Selection of region and meteorological stations**

   Under the proposed procedure, the initial step of studying regional profiles suggests a suitable choice of regions. To explore the history of drought, we recommend a wide-

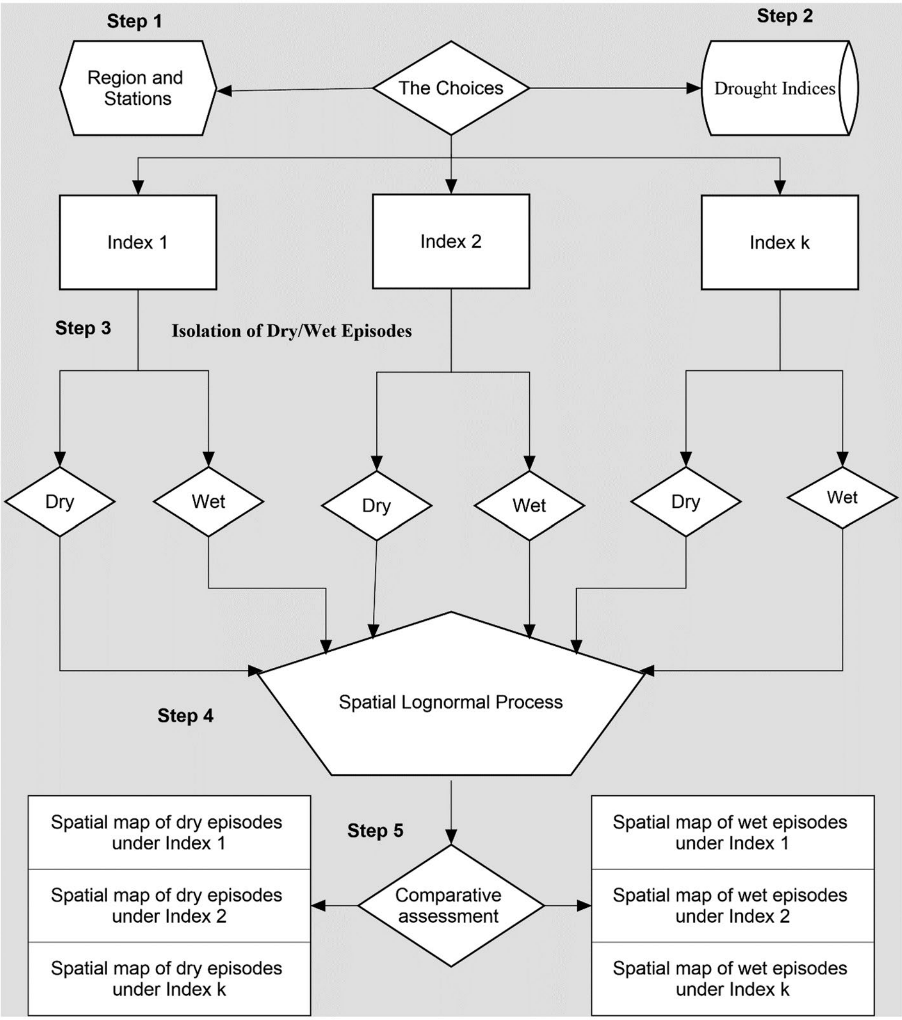

Fig. 1 Flowchart of the proposed procedure

ranging area consisting of several meteorological observatories with long-term time series. This step strengthens the reliability and scope of spatiotemporal distributions of Dry/Wet episodes.

2. **Selection of appropriate drought measures**

   This step focuses on the selection of comparable drought measures. In past research, many authors have provided numerous tools for the characterization of drought hazards under certain circumstances. However, the appropriate choice of drought monitoring tools increases the accuracy and reliability of drought assessment at a regional level.

3. **Isolation of Dry/Wet episodes**

   This step suggests the isolative analysis of Dry/Wet episodes under the selected drought measure for all stations in the area of interest. Further, the study presents an independent analysis of the Dry/Wet episodes in various time scales.

4. **The suitable choice of mapping spatial predictive distribution of Dry/Wet episodes**

   By considering spatial correlation in the count data of Dry/Wet episode, this step suggests a spatial Poisson log-normal model to infer the spatial predictive distributions of Dry/Wet episodes.

5. **Exploring regional profiles and spatiotemporal comparison**

   This step illustrates the explorative assessment of the historical profiles of Dry/Wet episodes at the regional level using spatial predictive maps under the spatial Poisson Log-Normal model. In addition, this step stimulates a comparative way to infer the intensities of Dry/Wet prone areas in spatial predictive maps.

3 Material and Methods

3.1 Data and Study Area

Along the same line as Ali et al. (2020), this research is based on fifty-two meteorological stations located in various parts of Pakistan. Figure 2 shows the geographical locations of the meteorological stations in Pakistan. Like other regions, Pakistan is also experiencing adverse global warming and climate change impacts. Numerous authors have reported the declining trend in rainfall in the northern areas. In the past few decades, drought has become one of the major problems for agriculture, energy, human life, and the ecosystem. Consequently, the country is facing several challenges regarding power crises, the decline in agricultural production and forestry, etc.

For this research, time-series data from 1970 to 2016 of precipitation and temperature are collected from the Pakistan Meteorological Department (PMD). The source and quality of the data have been described in Ali et al. (2019).

3.2 Standardized Drought Indices (SDIs) as a Drought Measures

Several multi-scalar drought indices have been developed to report drought severity for specific regions. However, each index has its own merits and demerits. Therefore,

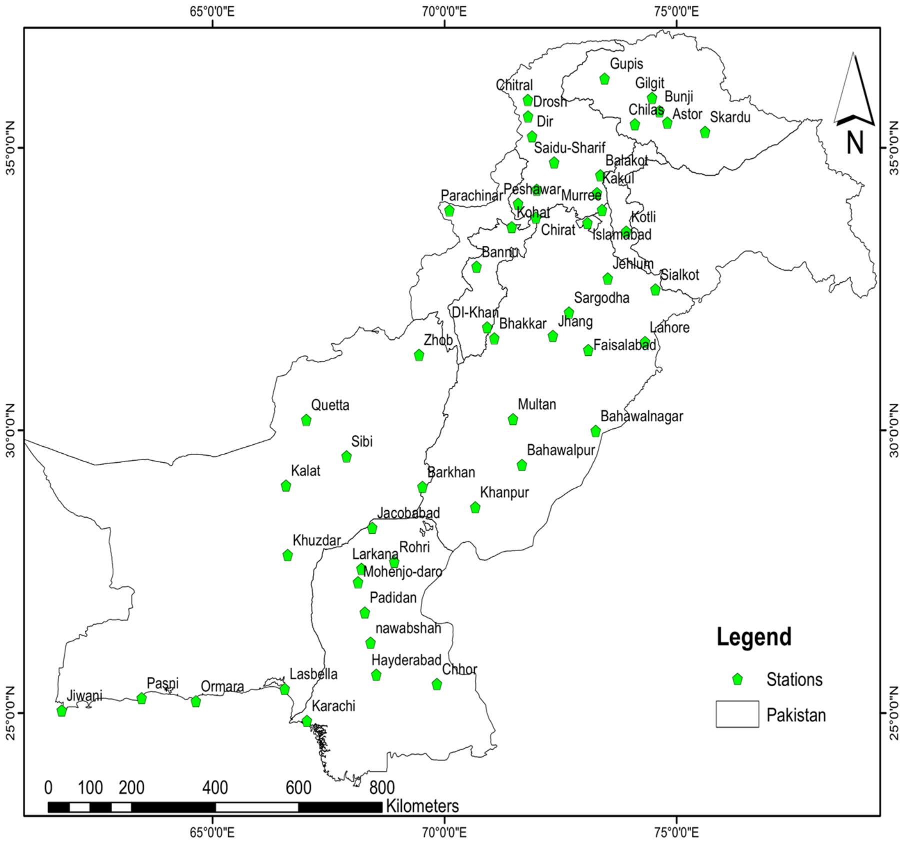

Fig. 2 Selected meteorological stations of Pakistan

the selection of drought indices for monitoring drought depends on the accessibility of data and the nature of the geographical region (Stagge et al. 2015). Under the rationale of our proposal, we have considered the three most commonly used SDIs, namely SPI, SPEI, and SPTI. SPI (McKee et al. 1993) is a simple, multi-scaler and one of the most commonly used drought monitoring indices (Montaseri et al. 2017). The methodology of SPI is based on the cumulative data of precipitation only. Vicente-Serrano et al. (2010) developed the SPEI drought index to assess and quantify drought. SPEI is a multi-scaler drought indicator like SPI. Unlike SPI, SPEI is a multivariate drought index that considers the impact of temperature and precipitation. The estimation procedure of SPEI is the same as of SPI. Ali et al. (2017) developed the Standardized Precipitation Temperature Index (SPTI) to monitor drought events in hot and cold climatic zones. SPTI is a multi-scaler drought index, and its estimation procedure is analogous to SPI and SPEI. Following McKee et al. (1993) and Vicente-Serrano et al. (2010) classification criteria, the resulting categories of each index are classified in Table 1. These drought indices are statistically comparable and have the ability to characterize drought on various time scales (Ali et al. 2019). Hence, the selection of these three indices is rationally valid.

Table 1 Drought classification criteria

SPI, SPEI, and SPTI	Classes	Major categories
≥ 2	Extremely Wet (EW)	Wet
1.5 *to* 1.99	Severely Wet (SW)	
1 *to* 1.49	Moderate Wet (MW)	
.99 *to* $-.99$	Near Normal (NN)	Normal
-1 *to* -1.49	Moderate Drought (MD)	Dry
-1.5 *to* -1.99	Severely Drought (SD)	
≤ -2	Extreme Drought (ED)	

4 Spatial Exploratory Analysis

4.1 Moran's *I*

The current study employed Moran's *I* coefficient of spatial autocorrelation for observing the spatial patterns of the geographical study area. Moran's *I* coefficient evaluates whether the observed patterns of the geographical region are clustered, dispersed, or random. This is the oldest method and is still helpful for exploring patterns of spatially scattered units. The mathematical formula of Moran's *I* coefficient is as follows:

$$I = \frac{n}{w_\circ} \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}(C_i - \overline{C})(C_j - \overline{C})}{\sum_{i=1}^{n} (C_i - \overline{C})}$$

where Ci is the total number of counts of historical events at a particular location, w_{ij} shows the weights between observation i and j, and W_0 is the sum of all weights. Mathematically, the following equation shows the formula of W_0:

$$W_\circ = \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}$$

Positive values of the Moran's *I* coefficients indicate clustered patterns, while a negative value shows that there is a dispersed pattern. However, values nearest to zero indicate the random pattern. The current study employed *vignettes* (Leisch 2003) package of R to estimate the values of the Moran's *I* coefficients. A separate analysis of historical spatially observed counts of Dry and Wet conditions has not been conducted in the past research.

4.2 Tjostheim's Coefficient and Modified t-test

Comparative analysis of drought indices at a spatial scale using traditional correlation analysis is inappropriate. This research employed Tjostheim's Coefficient (Tjøstheim 1978) and a modified t-test (Dutilleul et al. 1993) to check and test the association between each drought index. Tjostheim's coefficient measures the association between two stochastic processes observed over space. This paper compares the total counts of each Wet and Dry spell determined by each drought index under Tjostheim's Coefficient and modified t-test. In computation, *SpatialPack* (Osorio et al. 2012) R package is employed to compute and test the spatial association between SPI, SPEI, and SPTI at their respective scales.

4.3 Spatial Poisson-Lognormal Model

This paper explores the predictive distributions of the Wet and Dry categories of SPI, SPEI, and SPTI under the spatial Poisson log-normal model. A short description of the spatial Poisson log-normal model is as follows:

Let the vectors x_i and y_i contains spatial reference points. Further, the parallel non-Gaussian vector w_i consists on the quantitative data of Dry/Wet episodes. In this setting, Diggle et al. (1998) have suggested a class generalized linear spatial model (GLSM). GLSM is helpful for modeling spatial count data sets (Zhang 2002). Several studies used the GLSM model for modeling spatial count data in various disciplines. Royle and Wikle (2005) used spectral parameterization for the spatially varying mean of a Poisson model for mapping avian count data. Wakefield (2006) has used a generalized linear model for disease mapping in the presence of spatial autocorrelation in esophageal cancer incidence data. They suggested that the GLSM approach of modeling and mapping spatial counts of disease incidence is efficient and robust for inference. However, there exists a gap in the literature for mapping the spatial distribution of drought severity (Dry/Wet counts) conditions. The current study employed *geo-count* and *geoRglm* R packages to practically implement the spatial Poisson log-normal model under the MCMC algorithm.

5 Results

5.1 Calculation of Drought Indices

This study computed three standardized drought indices, namely SPI, SPEI, and SPTI, for fifty-two meteorological stations in Pakistan. In the part of Fig. 3A, C, and E, temporal behaviors of SPI, SPE, and SPTI are shown for Islamabad, Chilas, and Astor observatories. At the same time, the inner structures of each series are shown in its right parts B, D, and F parts of Fig. 3. These scatter plots show the significant dispersed behavior of SPEI from SPI and SPTI in all stations. Moreover, the distribution of SPEI varies from station to station.

However, the scatter plots and empirical distribution of SPI and SPTI are analogous in all the stations. We found that SPI and SPTI are strongly correlated in all the meteorological stations. Though, SPEI has had a relatively small correlation with SPI and SPTI. The reason is that; the computation methodologies of each index are different, and each model's error always exists. However, this correlation does not necessarily hold among each drought index in classified categories of drought classes. Further, a good probability model may exist beyond the candidate list in the selected model.

5.2 Spatial Exploratory Analysis

This section demonstrates the results associated with the spatial pattern Dry and Wet counts. Summary statistics for each index are given in Table 2. The observed Moran's I values for SPI-3, and SPEI-6 are positive. The p-value is less than the specified level of

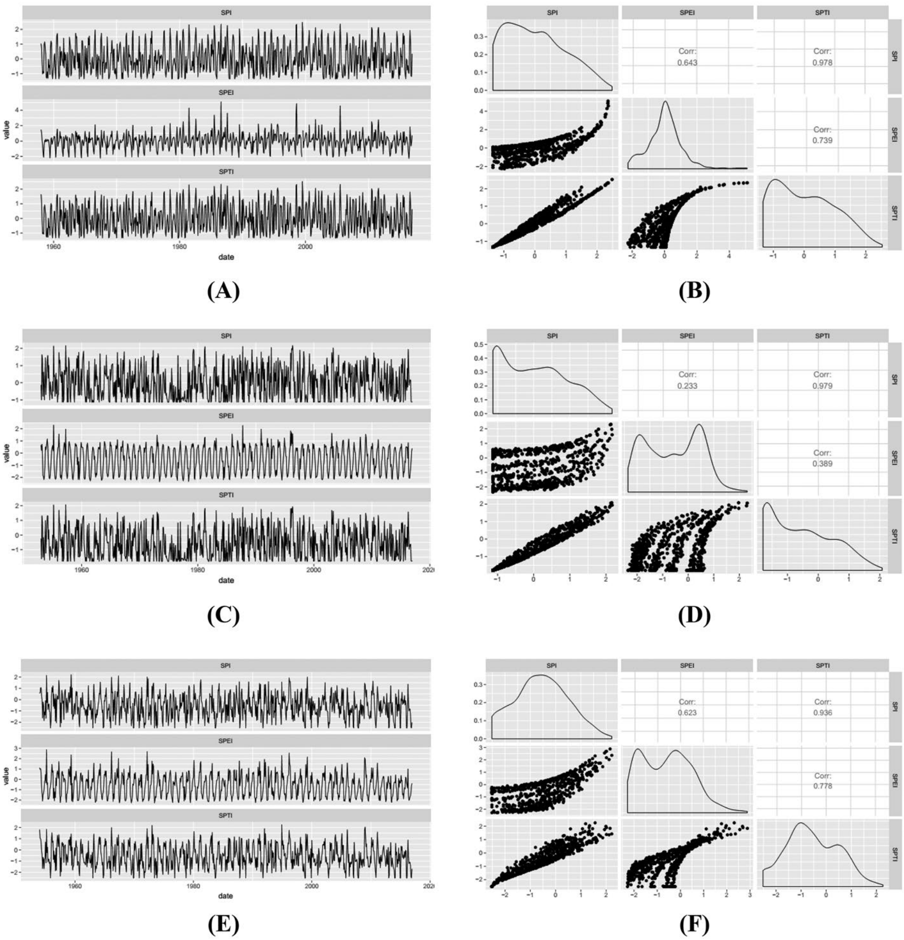

Fig. 3 Temporal behavior and relationship between SPI, SPEI and SPTI

significance in both Wet and Dry prevalence data, which indicates that the observed pattern of the geographical region is clustered. Besides this, SPTI-3, SPI-6, SPEI- 6, SPI-12, SPEI-12, and SPEI-12, negative values of Moran's I indicate a dispersed pattern of the observed geographical region, but all these patterns are insignificant. However, clustered and dispersed patterns are observed for Dry and Wet spells of frequency counts of SPEI-3, respectively.

Moreover, for SPTI-3 and SPI-6, insignificant random patterns are observed. Except for the wet spell of SPTI-9, which shows dispersed behavior of geographical location, a nine-month time scale possesses positive values in both Dry and Wet spells. Conclusively, varying patterns of observed geographical regions suggest choosing appropriate statistical and geospatial techniques for each index.

In Table 3, numerical estimates of spatial association among each time scale for Dry classes are given. Although, all the time scales have a significant positive spatial association. However, a strong positive correlation is found only in SPI-12 and SPTI-12. This

Table 2 Spatial pattern of each drought category

Index	Drought Category	Moran's I	Standard Deviation	P-value	Pattern
SPI-3	Dry	0.0554	0.0353	0.0351	Clustered
	Wet	0.0949	0.0349	0.0011	Clustered
SPEI-3	Dry	–0.0228	0.0348	0.9092	Dispersed
	Wet	0.0659	0.0335	0.0114	Clustered
SPTI-3	Dry	0.0089	0.0341	0.4154	Random
	Wet	0.0221	0.0343	0.2318	Random
SPI-6	Dry	0.0335	0.0353	0.1382	Random
	Wet	0.0445	0.0347	0.0680	Random
SPEI-6	Dry	0.0608	0.0352	0.0238	Clustered
	Wet	0.0866	0.0344	0.0022	Clustered
SPTI-6	Dry	–0.0062	0.0352	0.7193	Dispersed
	Wet	–0.0094	0.0350	0.7874	Dispersed
SPI-9	Dry	0.1044	0.0351	0.0004	Clustered
	Wet	0.0393	0.0349	0.0954	Random
SPEI-9	Dry	0.0894	0.0351	0.0020	Clustered
	Wet	0.0389	0.0349	0.0976	Random
SPTI-9	Dry	0.0069	0.0345	0.4553	Random
	Wet	–0.0236	0.0282	0.8672	Dispersed
SPI-12	Dry	–0.0560	0.0352	0.2925	Dispersed
	Wet	–0.0107	0.0350	0.8148	Dispersed
SPEI-12	Dry	–0.0475	0.0352	0.4160	Dispersed
	Wet	–0.0032	0.0350	0.6548	Dispersed
SPTI-12	Dry	–0.0482	0.0352	0.4037	Dispersed
	Wet	–0.0059	0.0349	0.7104	Dispersed

suggests using SPTI-12 as an alternative to SPI-12 at a spatial scale. Moreover, SPI-3 and SPEI-3 in Dry and Wet spells have weak positive associations. Similarly, a low correlation ($r = 0.3718$) between SPEI-3 and SPTI-3 in Wet spells (see Table 3) have observed accordingly. These findings recommend making separate analyses for each time scale of the three multi-scaler drought indices.

The following subsection forms spatial predictive distribution for both Wet and Dry spells of SPI, SPEI, and SPTI at 3,6, 9, and 12-month time scales using the spatial Poisson model.

5.3 Spatial Predictive Distributions

Figure 4 illustrates the spatial predictive distribution of Dry and Wet classes of SPI drought indices at 3, 6, and 12-time scales where, relative to the Wet classes, the frequency of the Dry classes is significantly high. This is because, in the climate of Pakistan, only two or three months, which includes July, August, and September, receive large amounts of precipitation (Ali et al. 2009). Moreover, the spatial predictive distribution of SPI-3 is

Table 3 Spatial correlation among drought indices

	Drought Indices		Spatial Correlation	F-statistics	*P*-value
Dry Classes	SPI-3	SPEI3	0.4691	10.103	0.0030
		SPTI-3	0.5831	26.079	0.0000
	SPI-6	SPEI-6	0.6740	28.636	0.0000
		SPTI-6	0.7582	56.864	0.0000
	SPI-9	SPEI-9	0.9873	1016.80	0.0000
		SPTI-9	0.3840	7.8974	0.0073
	SPI-12	SPEI-12	0.7282	59.134	0.0000
		SPTI-12	0.9456	366.72	0.0000
	SPEI-3	SPTI-3	0.4360	11.493	0.0014
	SPEI-6	SPTI-6	0.5504	21.143	0.0000
	SPEI-9	SPTI-9	0.3876	8.1086	0.0066
	SPEI-12	SPTI-12	0.7974	76.118	0.0030
Wet Classes	SPI-3	SPEI3	0.4235	8.783	0.0051
		SPTI-3	0.8146	85.210	0.0000
	SPI-6	SPEI-6	0.6456	24.897	0.0000
		SPTI-6	0.7887	56.752	0.0000
	SPI-9	SPEI-9	0.9792	676.960	0.0000
		SPTI-9	0.2721	4.1119	0.0478
	SPI-12	SPEI-12	0.7618	53.557	0.0000
		SPTI-12	0.9060	197.497	0.0000
	SPEI-3	SPTI-3	0.3718	7.642	0.0000
	SPEI-6	SPTI-6	0.2634	3.5535	**0.0655**
	SPEI-9	SPTI-9	0.6180	22.615	0.0000
	SPEI-12	SPTI-12	0.7634	61.281	0.0000

relatively consistent with the climatology of the stations. Analogous to SPI-3, the spatial distribution of both Wet and Dry spells of SPI-6 and SPI-9 behave equitably. However, a little discrimination can be found in SPI-12 where the frequency of Wet and Dry episodes is inconsistent at a spatial scale. The segregated patterns of the historical counts of both Dry and Wet conditions at various time scales of SPI spell explore some discrepancies in their spatial prevalence.

Additionally, to check the effect of evaporation on segregated spatial patterns of both Dry and Wet spells, the spatial Poisson model is applied to the historical counts of SPEI at 3, 6, 9, and 12-month time scales. Contrary to SPI, the spatial distribution of SPEI yields significantly different patterns of both Dry and Wet spells. A significant change in the spatial distribution of Dry and Wet prevalence is observed by including the effect of evaporation. The climate of Pakistan has four weather seasons. The coastal regions and the Arabian Sea are generally hot in summer and warm in winter weather. Moreover, the mountain areas, including the Northern areas, suffered from shallow temperatures throughout the year. This climate diversity is due to the high altitude of the region. Therefore, empirical findings can be cross-validated by seeing the spatial correlation between SPI and SPEI. Although, there is a significant spatial correlation between all the time scales of SPI and SPEI (see Table 3). However, the weak correlations (r = 0.4691 with P-value = 0.003, and

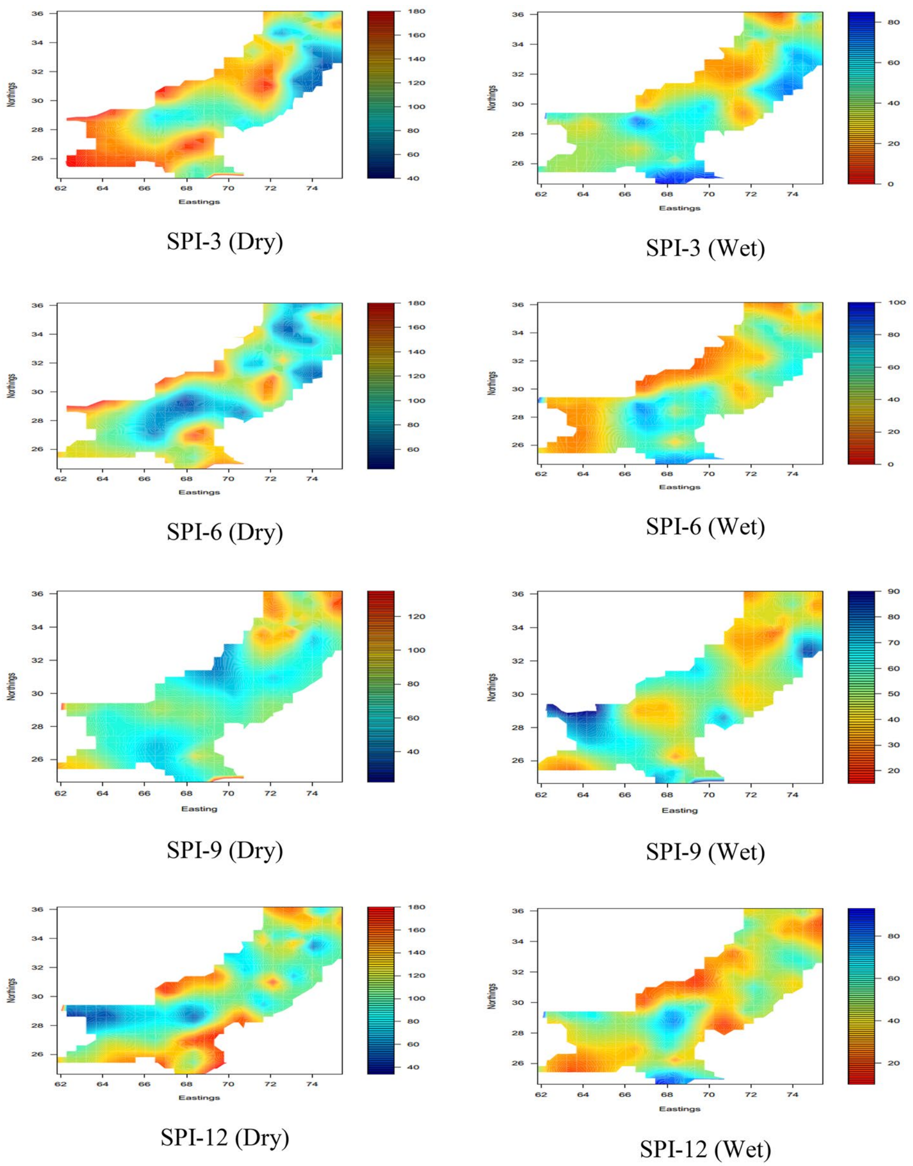

Fig. 4 Predictive distribution of Dry and Wet counts of SPI drought Index

r = 0.4231 with *p*-value = 0.0051) between SPI-3 and SPEI-3 in both Dry and Wet counts respectively, raise several questions to SPEI as a substitute of SPI.

This inconsistency may be caused by the role of evapotranspiration in the determination of SPEI, resulting in a large number of Wet spells in SPEI compared to SPI. Moreover, in the estimation procedure of SPEI, numerical estimates in low-temperature months and stations may cause disaggregation (Ali et al. 2017) in the temporal series of SPI and SPEI. We compared the segregated spatial distribution of Dry and Wet spells (see Figs. 4 and 5).

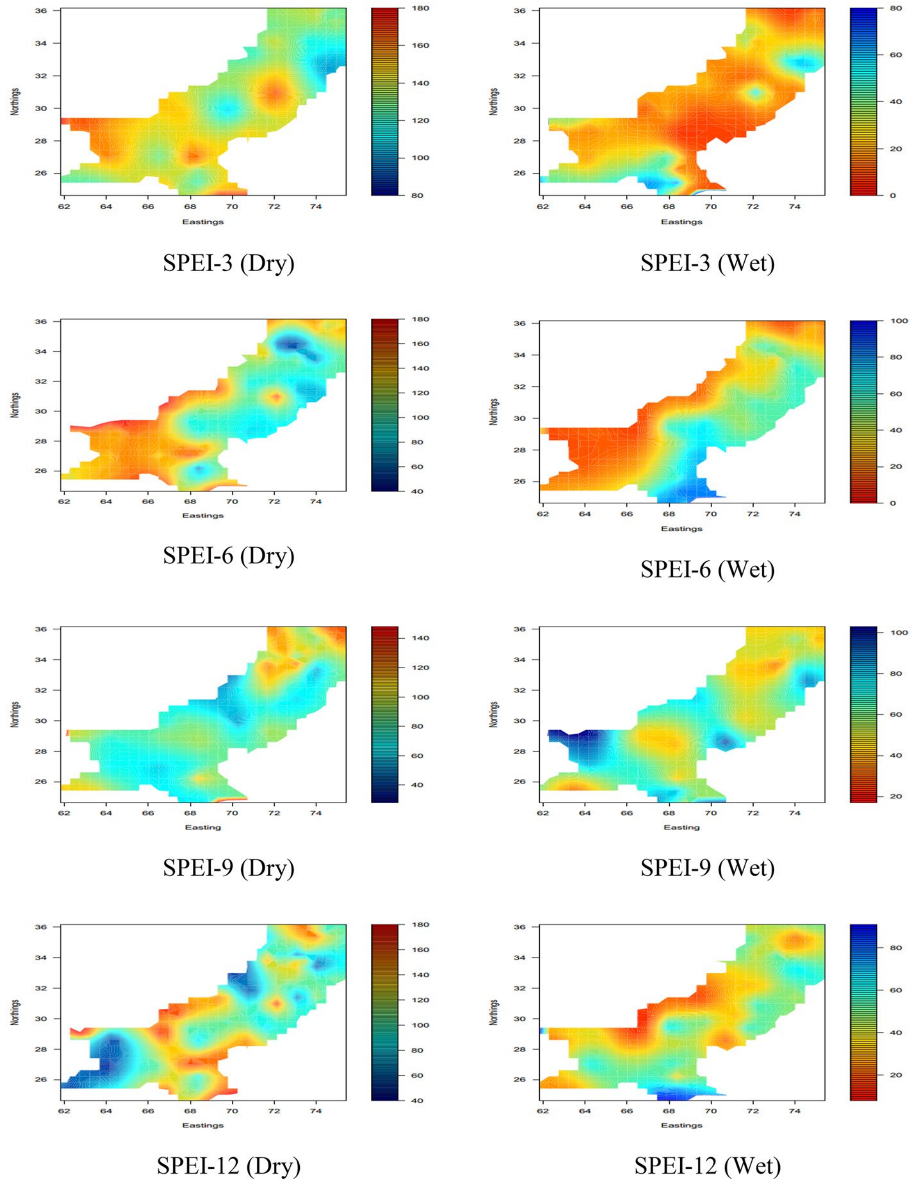

Fig. 5 Predictive distribution of Dry and Wet counts of SPEI drought Index

However, at six, nine, and twelve-month time scales, the spatial trend of SPEI is quite similar to SPI, and the frequencies of observed Dry categories are consistent.

Figure 6 shows the predictive distribution of Dry counts determined by SPTI. Including the direct role of temperature as suggested may lead to exploring more spatial variation according to the structure of the tropical geography of the regions. In Fig. 6, the observed historical frequencies of Wet/Dry conditions show inconsistencies

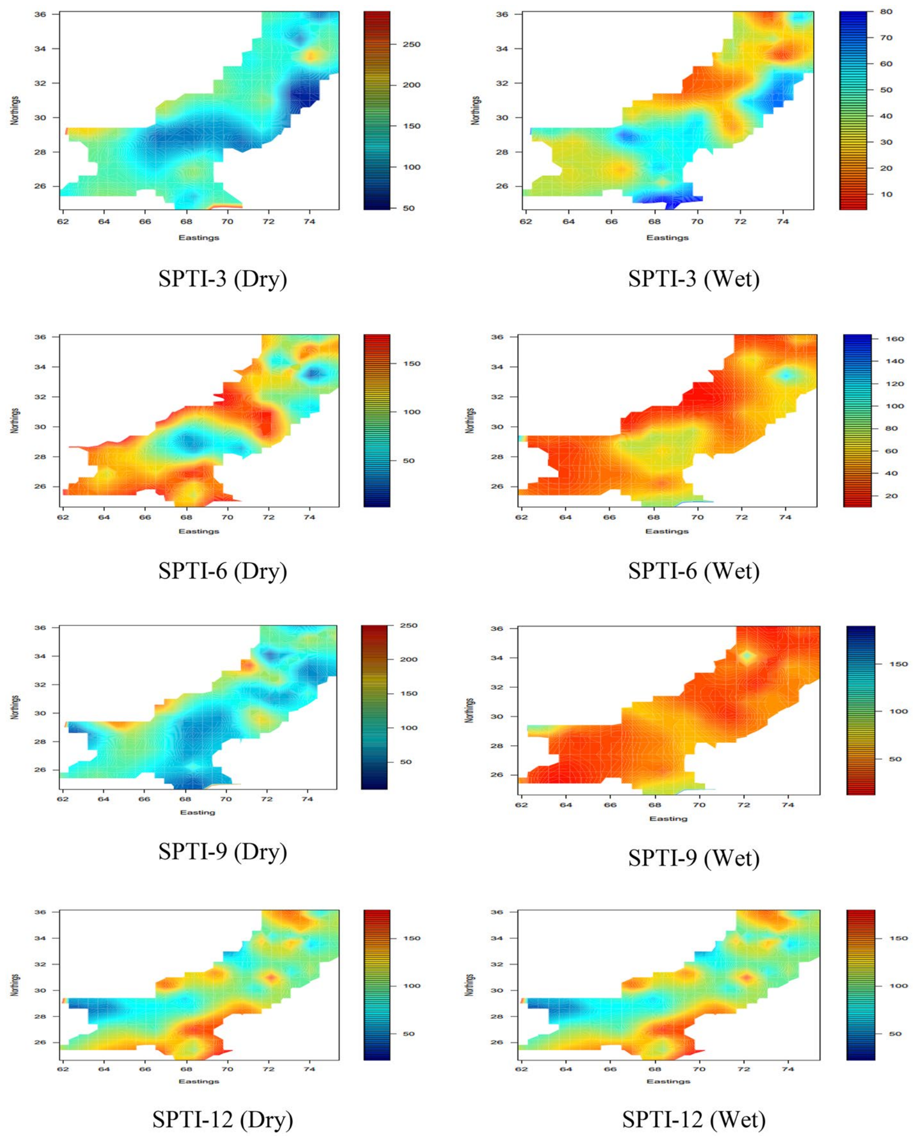

Fig. 6 Predictive distribution of Dry and Wet counts of SPTI drought Index

in the spatial maps. These inconsistencies show that spatial counts of the near-normal drought category are different from those observed in both SPI and SPEI. These results indicate that including the direct role of temperature largely contributes to the determination of Dry and Wet classes. Moreover, contrary to SPEI and SPI, this indication can be seen by assessing the spatial distribution of SPTI-6.

Overall, all three drought indices have a low frequency of the Wet spell compared to Dry spells in their historical counts. However, their spatial distributions of Dry and

Wet spells are quite different, indicating the use of more than one drought indices at a spatial scale for accurate and precise analysis of drought hazards.

6 Discussion

This paper has provided a new procedure to compare spatiotemporal patterns of Dry/Wet counts at a regional level. The proposed design incorporates three standardized drought indices at various time scales and the spatial Poisson log-normal model for fifty-two meteorological stations in Pakistan. Results of this research show that the segregated patterns of Dry/Wet counts are moderately consistent with the climatology of the region and significant differences among the spatial behavior of drought indices at various time scales. The inconsistency between the predictive distributions of SPI and SPTI specifies that including the direct role of temperature largely contributes to the determination of Dry and Wet classes. Therefore, we suggest simultaneous implications of more than one drought index at various time scales. Overall, all three drought indices have a low frequency of the Wet spell compared to Dry spells in their historical counts. These results indicate the prevalence of future drought.

Although the study gives a new comparative procedure to compare drought indices, the application has several limitations. While preparing predictive distribution (maps), we assume that all three indices have non-Gaussian features for Dry and Wet counts. Here, some numerical algorithms are required to test and fit the model (Jing and De Oliveira 2015). Therefore, the study proposed a Markov Chain Monte Carlo (MCMC) simulation algorithm for this task. However, the practical implementation of MCMC has several statistical challenges (Robert and Casella 1999; Brooks 1998).

Moreover, the prior distribution specification for GLSM is somehow an unexplored field. Many studies used non-informative priors, and it has been considered an analogy for hierarchical and non-hierarchal Gaussian models (Jing and De Oliveira 2015). Therefore, this study is based on non-informative priors.

Regardless of some statistical limitations, the outcomes of this research can guide hydrologists and meteorologists for better the use of and the selection of drought indices. In addition, the water resources and management authorities may use the predictive distribution of various drought types to report regional trends and spatial performance of different drought indices. Moreover, the study strengthens hydrological, meteorological, and climatological research by giving a new spatial comparative approach.

7 Conclusion

For effective and accurate drought mitigation policies, this research explores and compares regional profiles of spatiotemporal patterns of Dry/Wet episodes under various drought measures at different time scales. Under the spatial Poisson log-normal model, this article presented a new comprehensive procedure for comparing drought indices. The proposed method is based on the segregated spatial count data of Dry/Wet episodes to increase the reliability of comparative inferences. From this research, we have concluded, 1) the inconsistencies between the predictive distributions of SPI and SPTI specify that the direct role of temperature largely contributes to the determination of Dry/Wet classes, 2) the spatial distributions of Dry/Wet spells are different in each drought index. These inferences indicate that using only one drought index gives inaccurate and imprecise regional analysis of drought. Therefore, this research recommends multiple drought indices for regional drought analysis.

Author Contribution All the authors (Farman Ali, Zulfiqar Ali, Bing-Zhao Li, Sadia Qamar, Amna Nazeer, Saba Riaz, Muhammad Asif Khan, Rabia Fayyaz, Javeria Nawaz Abbasi) have an equal contribution.

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Declarations

Ethics Approval Not Applicable.

Consent to Participate Not Applicable.

Consent to Publish Not Applicable.

Competing Interests The authors declare no competing interests.

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